

SWORUP RANJAN NAYAK

AI/ML Engineer | Frontend Developer | Generative AI & Agentic Systems

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Professional Summary

Software Engineer with **3+ years** of experience building scalable **Healthcare Information Systems (HIS)**, **Electronic Medical Records (EMR)**, and real-time web applications using **React.js**, **JavaScript**, and **Redux Toolkit**, now specializing in **AI/ML and Agentic Systems**. Hands-on experience designing **multi-agent AI systems**, **Retrieval-Augmented Generation (RAG) pipelines**, and **LLM-powered applications** using **LangChain**, **LangGraph**, **Vector & Graph Databases**, and the **Model Context Protocol (MCP)**. Skilled at combining strong full-stack engineering fundamentals with applied Generative AI to build production-ready, real-time, and intelligent systems.

Technical Skills

- **AI / GenAI & Agentic Systems:** LangChain, LangGraph, Retrieval-Augmented Generation (RAG), Multi-Agent Systems, Model Context Protocol (MCP), Vector Databases (Pinecone, ChromaDB, Qdrant), Graph Databases (Neo4j), Speech-to-Speech (STS) AI, Prompt Engineering, LLM Context & Token Optimization, OpenAI API, Google Gemini API
- **Languages:** TypeScript, JavaScript (ES6+), Python, HTML5, CSS3
- **Frontend:** React.js, Next.js, Redux Toolkit, Context API, React Hooks, Styled Components, CSS Modules, React Testing Library
- **Backend:** Node.js, Express.js, REST APIs, JWT Authentication, WebSockets
- **Cloud & DevOps:** AWS (EC2, ECS, S3, IAM, API Gateway), Docker, GitHub Actions, CI/CD Pipelines
- **Tools:** Git, GitHub, Jira, Agile/Scrum, VS Code, AI-Assisted Development (ChatGPT, Claude, Gemini)

Education

B.Tech in Computer Science and Engineering — Silicon Institute of Technology, Bhubaneswar *CGPA: 9.1 / June 2023*

Professional Experience

Software Engineer (SDE-II) | Squbix Digital Pvt Ltd — Bhubaneswar, IN *Mar 2025 – Present*

- Led delivery of the IPD (In-Patient Department) module and dashboards used by 50+ clinicians, streamlining patient care workflows.
- Architected patient dashboards, vitals tracking, printable views, and 10+ tab-based clinical interfaces using React, Redux Toolkit, and CSS Modules.
- Designed a responsive grid layout system supporting nested widgets and patient-centric visualizations across multiple healthcare workflows.
- Strengthened application security through JWT authentication, RBAC, and permission mapping across 20+ user roles.
- Improved application performance through memoization, lazy loading, and dynamic imports, reducing load times by over 30%.
- Integrated 4+ EMR components including discharge summaries, clinical notes, vitals, and medication charts with backend APIs.
- Delivered Medication Management capabilities including **Infusion Therapy** and **Variable Dose Medication** workflows.
- Engineered a centralized **Notification Module** supporting real-time alerts across 4 major HIS modules.
- Enabled real-time synchronization using **WebSockets** across patient monitoring and clinical workflows.
- Established a configurable **Patient Alert System** supporting allergies, warnings, and critical patient information.
- Created shared component libraries containing 30+ reusable UI components, improving development consistency across applications.
- Partnered with QA and DevOps teams to streamline CI/CD pipelines using GitHub Actions and AWS ECS deployments.
- Designed **multi-agent AI systems** using **LangGraph**, improving clinical workflow automation efficiency by 35%.
- Built **multilingual conversational AI pipelines** converting unstructured patient-doctor interactions into structured EMR-ready data, reducing manual documentation effort by 50%.
- Optimized **LLM pipelines** with context management and prompt engineering, reducing token usage cost by 25%.

Software Engineer (SDE-I) | Squbix Digital Pvt Ltd — Bhubaneswar, IN *Sep 2023 – Mar 2025*

- Delivered a configurable **Healthcare Information System (HIS)** supporting forms, workflows, and role management through a no-code interface.
- Built an EMR platform supporting customizable patient workflows, reusable templates, and conditional rendering capabilities.
- Created modules for authentication, user management, employee onboarding, appointments, and patient visits.
- Engineered a drag-and-drop form engine with validations, conditional logic, and reusable field components, reducing form configuration effort by approximately 60%.
- Developed Nursing Station functionality for bed management, task delegation, shift tracking, and live operational updates.

AI / GenAI Projects

- **Github MCP** (Python, LLM Agents, FastMCP) — Command-line Github MCP server to connect to AI models and execute different Github Functions.
- **Data entry AI Agent** (LLM, FastAPI, Electron JS, PostgreSQL, OpenCV) — AI agent that screens and cleans handwritten documents to classify and extract data from the documents.
- **AI Portfolio Agent Chat** (LangChain, RAG, Vector DB, LLM API) — Conversational AI agent for an interactive portfolio site that answers visitor queries about projects, skills, and experience using RAG over embedded resume/project data.
- **RAG-Based AI Chatbot Framework** (LangChain, ChromaDB/FAISS, Embeddings) — Reusable Retrieval-Augmented Generation pipeline with document ingestion, chunking, and semantic search for grounded, low-hallucination chatbot responses.
- **Speech-to-Speech AI Coding Assistant** (STS AI, LLM, Real-time Audio) — Real-time speech-to-speech AI agent that accepts voice queries and returns spoken explanations and code suggestions for hands-free coding assistance.